8	(a)	(i)	The Moon is continuously attracted towards the Earth by gravitational force and orbits around the Earth. Explain why the Moon does not fall into the Earth.	
			.....	
			.....	
			.....	
			.....	[2]
		L2	The **gravitational force exerted by Earth on Moon provides for the centripetal force required for Moon's orbital motion**, **which only changes the direction of the Moon's linear velocity and not its magnitude**.	**B1** **B1**
		(ii)	Assuming Earth to be a perfect sphere, explain why the weight of a person measured on an electronic balance at the poles is greater than that measured at the equator.	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	[4]
		L3	Free body diagrams: Pole N_P F_g $F_c = 0$ Equator F_g N_E $F_c \neq 0$	

			N: Normal contact force exerted by electronic scale on person F_g: Gravitational force exerted by Earth on person **Electronic balance reads the normal contact force N exerted by the person on the electronic balance** (i.e. the apparent weight of the person). **At a pole, person does not undergo circular motion, hence centripetal force F_c equals zero. Thus N equals in magnitude to F_g.** **At the equator, person undergoes circular motion, hence F_c is non-zero. Thus N is lesser in magnitude than F_g.** **Since the Earth is assumed to be a perfect sphere, its radius at the poles and equator are equal, hence F_g at both locations are assumed equal. This implies that N at the poles is greater than N at the equator** (and hence the apparent weight of the person is lesser at the equator).	**B1** **B1** **B1** **B1**
	(b)	The Moon has an orbital period of about 27.3 days. The mass of the Earth is 6.00×10^{24} kg and the mass of the Moon is 7.34×10^{22} kg.		
		(i)	Show that the distance between the centres of the Moon and the Earth is approximately 383 000 km.	
				[3]
		L2	**Gravitational force F_G exerted by Earth on Moon provides for the centripetal force F_C required for Moon's circular motion.** $F_G = F_C$ $G\frac{M_E M_M}{r^2} = M_M r\omega^2$ $G\frac{M_E}{r^2} = r(\frac{2\pi}{T})^2$ $(6.67 \times 10^{-11})\frac{6.00 \times 10^{24}}{r^2} = r(\frac{2\pi}{27.3 \times 24 \times 60 \times 60})^2$ $r = 383.49 \times 10^6\ m$ $r = 383\,000\ km\ (3\ s.f.)$	**B1** **B1** **B1** A0

(ii) Hence, calculate the distance *x*, from the centre of the Earth along the Earth-Moon line to the point Q where the resultant gravitational field strength due to the Earth and Moon is zero.

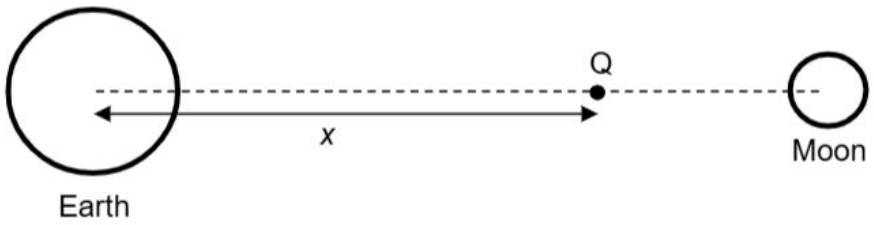


Figure 8.1 (not to scale)

x = ...m [2]

L1 As Q, $g_{net} = 0$, hence

$$|g_E| = |g_M|$$

$$G\frac{M_E}{x^2} = G\frac{M_M}{(383 \times 10^6 - x)^2}$$ **M1**

$$\frac{M_E}{M_M} = \frac{x^2}{(383 \times 10^6 - x)^2}$$

$$\frac{6.00 \times 10^{24}}{7.34 \times 10^{22}} = \frac{x^2}{(383 \times 10^6 - x)^2}$$

Solving,

$x = 3.4486 \times 10^8 =$ **3.45×10^8 m** **A1**

(iii) Calculate the gravitational potential at point Q.

gravitational potential at point Q = ...J kg^{-1} [2]

		L1	Gravitational potential at point Q $= -G\frac{M_E}{x} + (-G\frac{M_M}{(383\times10^6 - x)})$ $= -(6.67 \times 10^{-11})\left[\frac{6.00\times10^{24}}{3.4486\times10^8} + \frac{7.34\times10^{22}}{(383\times10^6 - 3.4486\times10^8)}\right]$ $= -1.2888 \times 10^6 =$ **- 1.29 x 10^6 J kg^{-1}**	M1 A1
		(iv)	The radius of the Earth is 6370 km and the radius of the Moon is 1740 km. Determine the minimum energy required to send a stationary capsule of mass 30 kg from the surface of the Earth to the surface of the Moon.	
			minimum energy = ……J	[4]
		L2	Gravitational potential at the **surface** of the **Earth** $= -G\frac{M_E}{R_E} + (-G\frac{M_M}{(383\times10^6 - R_E)})$ $= -(6.67 \times 10^{-11})\left[\frac{6.00\times10^{24}}{6370\times10^3} + \frac{7.34\times10^{22}}{383\times10^6 - 6370\times10^3}\right]$ $= -6.2839 \times 10^7$ J kg^{-1} For **minimum** energy requirement, capsule has **zero kinetic energy when it reaches point Q.** *[Recap: The point where g_{net} = 0 between Earth and Moon is also the point of maximum potential.]* Hence, the **minimum energy required** = m ΔΦ **= (30) (-1.2888 x 10^6 – (- 6.2839 x 10^7))** **= 1.85 x 10^9 J** *Note: Enough energy must be supplied for the capsule to overcome the maximum increase in GPE in moving from Earth to Moon, which is from Earth's surface to the neutral point. Once the capsule passes the neutral*	M1 B1 M1 A1

point, it will be experience a net gravitational force towards the Moon which accelerates it towards the Moon without the need for external energy to be supplied.

(c) A planet of radius R is isolated in space. It has no atmosphere, and its mass can be assumed to be concentrated at its centre.

A projectile is launched from the surface of the planet with a **minimum** kinetic energy ***E*** required for the projectile to **not** return to the planet.

On Figure 8.2, draw:

(i) a line, labelled **P**, to show how the gravitational potential energy of the projectile varies with distance from the centre of the planet as it moves from R to $4R$,

(ii) a line, labelled **K**, to show how the kinetic energy of the projectile varies with distance from the centre of the planet as it moves from R to $4R$.

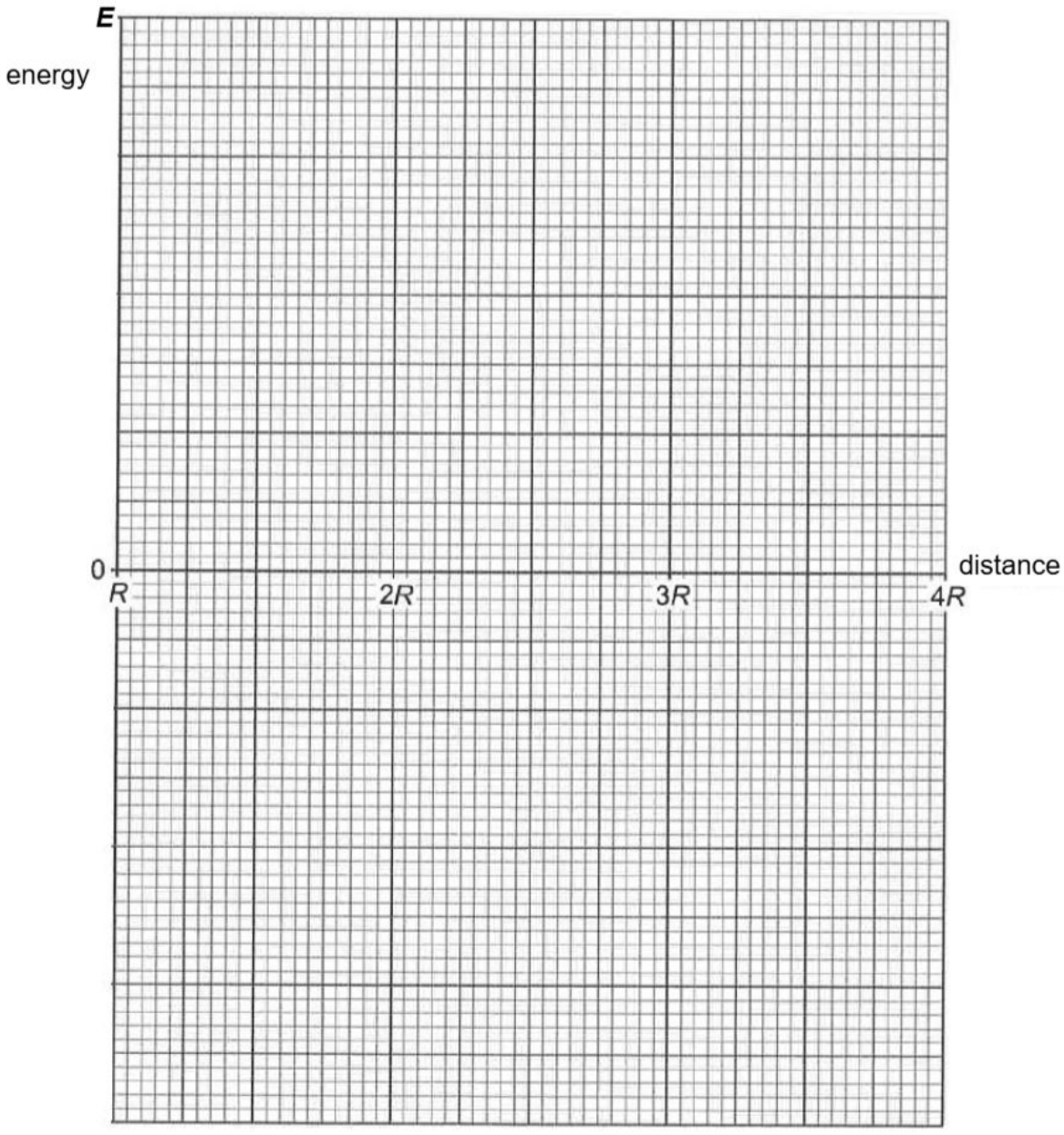


Figure 8.2 [3]

L2 *Q: What is the initial value of GPE?*

***Total energy is conserved**. Since $E_{T,\infty} = E_{K,\infty} + E_{P,\infty} = 0 + 0$, therefore **at every instant, $E_T = 0$**.*
*So, **$E_K + E_P = 0$ throughout**.*
*Thus, if $E_{K,\text{surface}} = E$, then **$E_{P,\text{surface}} = -E$**.*

Q: What is the variation of GPE with distance from centre of planet like?

$E_P = -\frac{GMm}{r}$, *where GMm remains constant.*

$\therefore E_P \propto \frac{1}{r}$ *and negative. (i.e. when r is doubled, E_P is halved, etc.)*

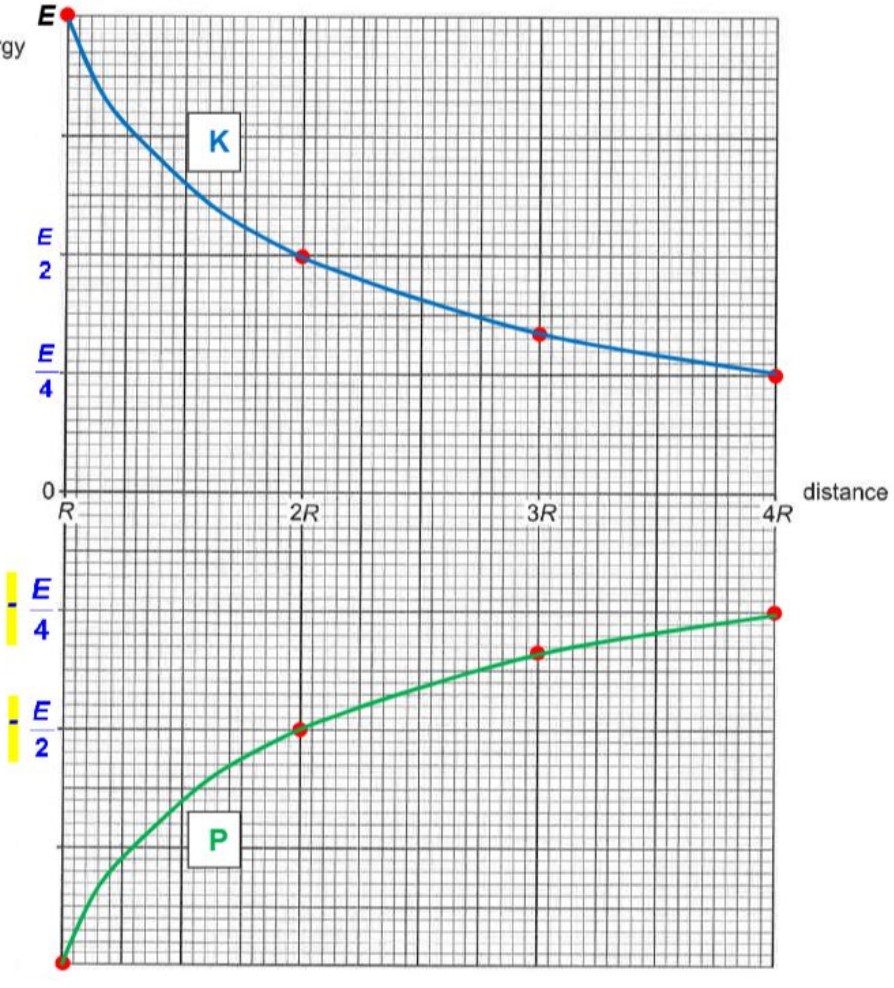
(i) *Tabulating,*

r	***E_P***	***Remarks***
R	$-E$	*Value of $E_{P,\text{surface}}$ found earlier.*
2R	$-\frac{E}{2}$	*20 small divisions vertically below horizontal axis*
3R	$-\frac{E}{3}$	*~13.3 small divisions vertically below horizontal axis*
4R	$-\frac{E}{4}$	*10 small divisions vertically below horizontal axis*

(ii) ***Since $E_T = 0$ throughout**, we can deduce the corresponding values of E_K at these instances. Tabulating,*

r	***E_K***
R	$+E$
2R	$+\frac{E}{2}$
3R	$+\frac{E}{3}$
4R	$+\frac{E}{4}$

E_P-r graph is a negative hyperbola; E_K-r graph is a positive hyperbola.
And the sum of both graphs at every point equals zero since $E_T = 0$.



B1

B1

B1

Marks scheme:

1 mark: Hyperbolic shape of both graphs, and, GPE negative while KE positive. Both graphs labelled as instructed.

1 mark: Knowledge that TE = 0 throughout.

1 mark: Precision of both graphs – GPE and KE values at R, 2R, 3R and 4R determined and plotted accurately to half-smallest division. [No penalty if y-axis not labelled, but points should be plotted accurately based on the E given.]

[Total: 20]

END OF PAPER 3